सी एस आई आर – राष्ट्रीय भौतिक प्रयोगशाला
CSIR-NATIONAL PHYSICAL LABORATORY
(वैज्ञानिक तथा औद्योगिक अनुसंधान परिषद्)
(Council of Scientific and Industrial Research)
(राष्ट्रीय मापिकी संस्थान (एनएमआई), सदस्य बीआईपीएस एवं हस्ताक्षरकर्ता सीआईपीएम - एमआरए)
(National Metrology Institute (NMI), Member BIPM and Signatory CIPM - MRA)
डॉ. के. एस. कृष्णन् मार्ग, नई दिल्ली-110012, भारत
Dr. K. S. Krishnan Marg, New Delhi-110012 INDIA
दूरभाष/Phone : +91-11-4560, 8441, 8610, 9447, फैक्स/Fax : 91-11-4560 8448
ई-मेल/E-mail : cfct@nplindia.org, वेबसाईट/Website : www.nplindia.org

अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE

A SET OF WEIGHTS

प्रमाण पत्र संख्या / Certificate No.

25050101/D1.01/C-025/amend

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pages
10.06.2025	10.06.2028	1	2

1. **Calibrated for** : Hi Tech Calibration Service Private Limited
No.209,VGP Nagar, Mugappair West,
Chennai - 600 037, Tamil Nadu , India

Customer's Ref. No : HT/LT/2025-26/013 Dated : 03.05.2025

2. **Description and identification of instrument**

10 kg to 1 g (17 Nos.)	:	Integral knob cylindrical stainless steel weights.
500 mg to 1 mg (12 Nos.)	:	Stainless steel wire weights.
Density (d)	:	(8000 ± 30) kg/m^3; k = 2
Make	:	LCGC
Serial No.	:	2025ZZ1 for (10kg), 2025AA1 for (5kg), 2025BB1 for (2kg), 2025BB2 for (.2kg), 2025CC1 for (1kg), 2025DD1 for (500g) & 2025211 for (200g to 1mg)

3. **Environmental conditions** : Temperature : (23.0 ± 1.5) °C
Relative humidity : (50 ± 10) %

[Change in temperature and relative humidity during the calibration were less than ± 0.3 °C per hour and ± 5.0 % per 4 hours respectively]

4. **Standard(s) used with associated Measurement uncertainty** : NPL transfer standard (Sub-Div.#1.01/M/TS/.S/57) of mass with uncertainty ± 0.053 mg: k=2.

5. **Metrological Traceability of Standard(s) used** : The standard(s) used for calibration is(are) traceable to the National Standard which realize the units of quantities according to the International System of Units (SI).

6. **Principle/Methodology of Calibration & Calibration Procedure number** : CSIR-NPL Calibration Procedures No. Sub-Div#1.01/Doc.3/CP#WT/M-02. Method of comparison with the CSIR-NPL transfer standard(s) using subdivision weighing. The reported mass value(s) is(are) the conventional mass value(s) (M_c) related to the true mass value (s) (M_t) by formula : $M_c = M_t \times [1 - 1.2 \times (1/d - 1/8\,000)]$.

अंशांकनकर्ता:
Calibrated by:
(D.C. SHARMA)

जाँचकर्ता:
Checked by:
(NIDHI SINGH)

प्रभारी वैज्ञानिक:
Scientist-in-charge:
(NIDHI SINGH)

जारीकर्ता:
Issued by: डॉ. श्रीनिवास राव रागम
Dr. Srinivasa Rao Ragam

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अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE

A SET OF WEIGHTS

प्रमाण पत्र संख्या / Certificate No.

25050101/D1.01/C-025/amend

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pages
10.06.2025	10.06.2028	2	2

7. Results :

Denomination	Box Id. No.	Mass Value (g)	Uncertainty (g)
10 kg	M/MV/W/10/04	9 999.998 8	± 0.001 6
5 kg	M/MV/W/5/08	4 999.999 0	± 0.000 8
2 kg	M/MV/W/2/16	1 999.999 6	± 0.000 3
2 kg	M/MV/W/2/15	2 000.000 2	± 0.000 3
1 kg	M/MV/W/1/08	999.999 952	± 0.000 160
500 g	M/MV/W/0.5/09	500.000 140	± 0.000 080
200 g	M/MV/WB/11	200.000 049	± 0.000 030
. 200 g	,,	200.000 067	± 0.000 030
100 g	,,	100.000 023	± 0.000 016
50 g	,,	50.000 016	± 0.000 010
20 g	,,	20.000 012	± 0.000 008
. 20 g	,,	20.000 011	± 0.000 008
10 g	,,	10.000 014	± 0.000 006
5 g	,,	5.000 008	± 0.000 005
2 g	,,	2.000 003	± 0.000 004
. 2 g	,,	2.000 002	± 0.000 004
1 g	,,	1.000 003	± 0.000 003
500 mg	,,	0.500 003	± 0.000 003
200 mg	,,	0.200 003	± 0.000 002
. 200 mg	,,	0.200 002	± 0.000 002
100 mg	,,	0.100 003	± 0.000 002
50 mg	,,	0.050 001	± 0.000 001
20 mg	,,	0.020 001	± 0.000 001
. 20 mg	,,	0.020 002	± 0.000 001
10 mg	,,	0.010 001	± 0.000 001
5 mg	,,	0.005 001	± 0.000 001
2 mg	,,	0.002 001	± 0.000 001
. 2 mg	,,	0.002 000	± 0.000 001
1 mg	,,	0.001 001	± 0.000 001

The reported uncertainty is at coverage factor k = 2 which corresponds to a coverage probability of approximately 95 % for a normal distribution.

8. Date(s) of Calibration : (04 to 05).06.2025

9. Remarks :
1. Mass value(s) of the weight(s) is(are) within the maximum errors permissible in E_1 accuracy class of weights as per OIML R 111-1 : 2004.
2. Dots(.) are used to distinguish the weights of same nominal value.
3. Density of weights are provided by customer.
4. This calibration certificate is issued on 30.09.2025 in lieu of the earlier certificate number 25050101/D1.01/C-025 dated 10.06.2025.

अंशांकनकर्ता:
Calibrated by:
(D.C. SHARMA)

जाँचकर्ता:
Checked by:
(NIDHI SINGH)

प्रभारी वैज्ञानिक:
Scientist-in-charge:
(NIDHI SINGH)

जारीकर्ता:
Issued by:
डॉ. श्रीनिवास राव रागम
Dr. Srinivasa Rao Ragam

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अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE
A SET OF WEIGHTS

प्रमाण पत्र संख्या / Certificate No.

25050101/D1.01/C-025

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pa ges
10.06.2025	10.06.2028	1	2

1. **Calibrated for** : Hi Tech Calibration Service Private Limited
No.209,VGP Nagar, Mugappair West,
Chennai - 600 037, Tamil Nadu , India

Customer's Ref. No.: HT/LT/2025-26/013 Dated : 03.05.2025

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Make	:	LCGC
Serial No.	:	2025ZZ1 for (10kg), 2025AA1 for (5kg), 2024BB1 for (2kg), 2025BB2 for (.2kg), 2024CC1 for (1kg), 2025DD1 for (500g) & 2024211 for (200g to 1mg)

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अंशांकनकर्ता:
Calibrated by:
(D.C. SHARMA)

जाँचकर्ता:
Checked by:
(NIDHI SINGH)

प्रभारी वैज्ञानिक:
Scientist-in-charge:
(NIDHI SINGH)

जारीकर्ता:
Issued by:
डॉ. श्रीनिवास राव रागम
Dr. Srinivasa Rao Ragam
National Physical Laboratory Director's Nominee New Delhi

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अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE
A SET OF WEIGHTS

प्रमाण पत्र संख्या / Certificate No.

25050101/D1.01/C-025

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pa ges
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The reported uncertainty is at coverage factor k = 2 which corresponds to a coverage probability of approximately 95 % for a normal distribution.

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Scientist-in-charge:
(NIDHI SINGH)

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National Physical Laboratory Director's Nominee New Delhi
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Dr. Srinivasa Rao Ragam